

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

Repeated DNS requests are being sent for the A record of `yummyrecipesforme.com`, and each request results in an ICMP error indicating that UDP port 53 is unreachable. Port 53 is used for DNS service. The most likely issue is that the DNS service is unavailable or that access to UDP port 53 is being blocked or misconfigured.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

At 13:24:32 Several customers of the client reported that they were unable to access the company website, `www.yummyrecipesforme.com`, and received a "destination port unreachable" error after waiting for the page to load. I attempted to access the website and received the same error. I loaded the network protocol analyzer tool, `tcpdump`, and attempted to access the webpage again while capturing the network traffic. The `tcpdump` analysis showed that UDP packets were sent to the DNS server on port 53. In response, ICMP packets returned the error message "udp port 53 unreachable," indicating that the DNS service could not be reached through UDP port 53. The likely cause is that the DNS service is unavailable or UDP port 53 is being blocked, preventing DNS requests from reaching the service. The next step in the investigation would be to determine whether the DNS service is down or whether UDP traffic to port 53 is being blocked by a firewall or other network configuration.